

Project Name: PlayOps — KK Wagh Sports Portal	Target Milestone: 30% Overall Progress Stage
Institution: KK Wagh Institute of Engg. Education & Research	Evaluation Sprint: Sprint 0 & Sprint 1 (Weeks 1–2)
Technology Stack: Next.js 15, TypeScript, Supabase, Tailwind, shadcn/ui	Current Status: 30% Milestone Reached & Approved

30% ROADMAP COMPLETED			
MILESTONE STATUS 30% REACHED Foundation Phase Complete	COMPLETED TASKS 41 / 137 Tasks 100% of Phase 1 Done	DATABASE LAYER 12 Relational Tables Deployed with RLS Policies	CORE AUTH ENGINE 4 RBAC Roles JWT & Middleware Enforced

1. Executive Summary & Milestone Inception

The primary objective of the **30% project milestone** was establishing an enterprise-grade architectural foundation for PlayOps — the centralized sports operations portal for KK Wagh College of Engineering. This phase focused on zero-defect foundational engineering: creating the modern full-stack application shell with Next.js 15 App Router, designing and deploying the full relational database schema on PostgreSQL (Supabase) with Row-Level Security (RLS), building a production-ready authentication and Role-Based Access Control (RBAC) engine, and launching the institutional landing page. All planned tasks for this initial 30% stage have been completed, verified against specifications, and deployed.

Key Architectural Achievement

Architectural Foundation Highlight: The PostgreSQL schema deployed at this milestone anticipates all future tournament, match, and certificate requirements without requiring destructive schema modifications later. Complete Row-Level Security (RLS) policies prevent cross-tenant data leaks right from day one.

2. Scope of Work Completed in First 30% Roadmap

- The first 30% of PlayOps focuses on establishing foundational stability across four core vectors:
- Modern Developer Toolchain:** Next.js 15 App Router, TypeScript strict typing, Tailwind CSS v3 design tokens, and the complete shadcn/ui accessible component primitive system.
 - PostgreSQL Relational Schema:** 12 fully indexed tables with UUID primary keys, foreign constraints, automated triggers, and seed scripts for college sports and grounds.
 - Authentication & Role Security:** Supabase Auth with JWT token persistence, multi-role definitions (Admin, Organizer, Player, Viewer), and Next.js Edge route guard middleware.
 - Universal Application Shell:** Dynamic topbar, collapsible role-aware sidebar, mobile drawer, error boundaries, and an engaging college landing page.

2. Detailed Technical Breakdown of Completed Modules (0% to 30%)

2.1 System Architecture, Toolchain & UI Primitives

- **Next.js 15 App Router Architecture:** Scaffolded project directories segregating public marketing routes ((public)), authenticated player views (/dashboard), and administrative operations (/admin).
- **Design Tokens & Theming:** Implemented Tailwind CSS design variables aligning with KK Wagh's visual identity (Deep Navy, Royal Blue, Slate Gray, Emerald Green).
- **shadcn/ui Component Suite:** Installed, customized, and styled foundational UI components: Button, Input, Card, Dialog, Table, Tabs, Badge, DropdownMenu, Sheet, and Sonner toast notifications.

2.2 PostgreSQL Database Modeling (12 Core Relational Tables)

Designed and executed migration scripts deploying 12 fully indexed relational tables on Supabase PostgreSQL:

1. **profiles:** Central student/player and faculty credentials, contact, PRN, department, and academic year.
2. **sports:** Catalog of supported games (Cricket, Football, Volleyball, Kabaddi, Badminton, Chess) with rule presets.
3. **venues:** Campus sports infrastructure (Main Oval, Football Turf, Synthetic Courts) with capacity and status.
4. **teams & team_members:** Departmental sports rosters, captain designations, and membership verification.
5. **tournaments & tournament_teams:** Multi-stage competition structures, formats, seedings, and registration entries.
6. **matches & match_events:** Game scheduling, venue allocation, real-time score events (goals, wickets, cards).
7. **points_table:** Dynamic standings storing Played, Won, Lost, Drawn, Points, and Net Run Rate / Goal Difference.
8. **notifications & certificates:** In-app alerts, broadcast logs, and digital merit/participation credentials.

2.3 Authentication, Session Management & Multi-Role Access Control (RBAC)

- **Supabase Auth Integration:** Secure email/password authentication with JWT token persistence and session cookies.
- **Multi-Role Hierarchy:** Defined four explicit authorization tiers: *Admin* (Full management), *Organizer* (Match/Tournament control), *Player* (Roster & registration), and *Viewer* (Public spectator).
- **Edge Route Protection Middleware:** Implemented src/middleware.ts to automatically refresh auth sessions, intercept unauthenticated requests to protected dashboards, and redirect unauthorized roles away from administrative consoles.

2.4 Application Shell Layout & Public Landing Experience

- **Master Layout Shell:** Responsive sticky navbar with authenticated user dropdown, mobile drawer, and institutional footer.
- **Collapsible Admin Sidebar:** Dynamic sidebar menu adjusting links based on logged-in user permissions.
- **Public Landing Portal:** Engaging homepage featuring hero banner, live tournament counters, quick-action CTA buttons, and college sports highlights.

3. Master Deliverables Verification Matrix — First 30% Scope

Task ID	Deliverable / Component	Category	Priority	Status
TSK-01	Next.js 15 + TypeScript Scaffolding	Setup	P0 Must-have	[VERIFIED]
TSK-02	Tailwind CSS + shadcn/ui Theme Setup	Setup	P0 Must-have	[VERIFIED]
TSK-03	Supabase Client & Server SDK Config	Backend	P0 Must-have	[VERIFIED]
TSK-04	Auth Middleware & Route Guarding	Security	P0 Must-have	[VERIFIED]
TSK-05	12 Relational DB Tables Migrations	Database	P0 Must-have	[VERIFIED]
TSK-06	Row-Level Security (RLS) Policies	Security	P0 Must-have	[VERIFIED]
TSK-07	User Profile Auto-Sync DB Trigger	Database	P1 High	[VERIFIED]
TSK-08	Sports & Venues Seed Data Scripts	Database	P1 High	[VERIFIED]
TSK-09	Registration & Login Pages	Auth UI	P0 Must-have	[VERIFIED]
TSK-10	Password Reset & Recovery Flow	Auth UI	P1 High	[VERIFIED]
TSK-11	Role-Based Access Control Engine	Security	P0 Must-have	[VERIFIED]
TSK-12	Responsive Topbar & User Nav	UI Shell	P0 Must-have	[VERIFIED]
TSK-13	Collapsible Navigation Sidebar	UI Shell	P0 Must-have	[VERIFIED]
TSK-14	Public Landing Page Hero & Stats	Frontend	P0 Must-have	[VERIFIED]
TSK-15	Global Error & Loading Boundaries	Frontend	P1 High	[VERIFIED]

4. Bridge to 50% Milestone — Upcoming Roadmap

With the solid 30% architectural base established, development proceeds directly into **Phase 2 (Core Management)** and **Phase 3 (Teams & Tournaments)** to reach the 50% milestone:

- 1. **Sports & Venue CRUD:** Implementing interactive administrative interfaces for campus facilities.
- 2. **Player Registration & QR Code Pass:** Generating unique cryptographic QR passes for all registered athletes.
- 3. **Team Creation & Roster Verification:** Department-wise team formation and captain transfers.
- 4. **Automated Fixture Algorithms:** Single-elimination knockout bracket generation and round-robin scheduling.

Formal Verification & Institutional Acceptance

Prepared By: Lead Full-Stack Engineering Team PlayOps Architecture Group	Verified By: Technical Project Lead & QA Dept. of Computer Engineering	Approved By: Director of Physical Education & Sports KK Wagh Institute of Eng. Edu. & Res.
Signature: _____ Date: 05-Oct-2026	Signature: _____ Date: 05-Oct-2026	Signature: _____ Date: 05-Oct-2026